

KRYLOV METHODS FOR THE MARKOWITZ PORTFOLIO

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Abstract. The Markowitz mean-variance portfolio problem, subject to general linear equality and inequality constraints, admits a symmetric indefinite KKT saddle-point formulation that MINRES [14] and a constraint-eliminated variant of CG [11] approximate efficiently—without ever forming the sample covariance matrix. Extending the minimum-variance Krylov framework of [17], the present paper identifies four generalisations required for the full Markowitz setting: general linear equality constraints encoded in a matrix A (factor-exposure targets, sector-weight targets); general linear inequality constraints $C^\top w \leq d$ (e.g. sector caps) handled by an active-set loop; a return tilt $\rho \mu^\top w$; and a Ledoit-Wolf ridge $\alpha(\|X\|_F^2/n)\|w\|^2$. In each case the KKT saddle-point structure is preserved and both solvers apply without modification. The saddle-point operator is implemented matrix-free as a `LinearOperator`, avoiding the $O(Tn^2)$ formation cost and $O(n^2)$ storage of $X^\top X$; the active-set loop is decoupled from the inner solver via a function-pointer interface, so the same loop drives the direct KKT, MINRES, and conjugate gradient [11] solvers uniformly.

Numerical experiments on synthetic data and on a real S&P 500 universe demonstrate order-of-magnitude speedups over open-source interior-point solvers for single-point and efficient-frontier solves, with and without Ledoit-Wolf shrinkage.

1. Introduction. Mean-variance portfolio optimisation, introduced by Markowitz [13], remains the dominant framework for quantitative asset allocation. In practice, portfolios are subject to a range of linear constraints beyond the basic budget requirement: sector caps, factor-exposure limits, individual position bounds, and long-only restrictions are routinely imposed. Solving the resulting constrained quadratic programme efficiently is therefore of direct practical importance, especially as the number of assets n and the frequency of re-optimisation grow.

The companion paper [17] showed that the long-only *minimum variance* portfolio reduces to a symmetric indefinite KKT saddle-point system, and that MINRES [14] and a constraint-eliminated variant of CG [11] approximate its solution efficiently, yielding large speedups over CVXPY [6] on standard benchmarks. The present paper extends this framework to the full *Markowitz mean-variance* problem. Four generalisations are required: (i) the single budget constraint is replaced by a general equality constraint matrix A ; (ii) general linear inequalities $C^\top w \leq d$ are handled by an active-set loop that promotes binding constraints to equalities; (iii) a linear return tilt $\rho \mu^\top w$ is added to the objective; and (iv) a Ledoit-Wolf [12] ridge $\alpha(\|X\|_F^2/n)\|w\|^2$, with $\alpha \in [0, 1]$. In each case the KKT saddle-point structure is preserved and MINRES applies without modification.

Krylov subspace methods for portfolio optimisation have previously been proposed by [1], who project the sample covariance matrix onto a Krylov subspace and solve the resulting low-dimensional problem, requiring $X^\top X$ to be formed explicitly; the present approach instead exploits the KKT saddle-point structure of the first-order optimality conditions, applying MINRES to the indefinite system without ever forming $X^\top X$.

A central contribution of the paper is a modular software architecture that separates concerns cleanly. The full problem specification—returns matrix, constraints, shrinkage parameters, and return-tilt weight—is encapsulated in a single frozen `Problem` dataclass (§2). The saddle-point operator is implemented matrix-free as a `scipy.sparse.linalg.LinearOperator` (§5), avoiding the $O(Tn^2)$ formation cost and $O(n^2)$ storage of $X^\top X$. The active-set loop is decoupled from the inner linear

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solver via a function-pointer interface (§4), so the same loop drives the direct KKT solve, MINRES, and conjugate gradient [11] solvers without code duplication.

The remainder of the paper is organised as follows. Section 2 states the problem and introduces the **Problem** dataclass. Section 3 derives the KKT saddle-point system, establishes nonsingularity and optimality, and analyses MINRES convergence via the spectral structure of the saddle-point matrix. Section 4 describes the active-set method. Section 5 presents the four solvers and their implementation, including a discussion of the fundamental differences from CVXPY and clarabel. Section 6 reports numerical experiments.

2. Problem Formulation. Let $X \in \mathbb{R}^{T \times n}$ be the demeaned (column-wise) daily return matrix for n assets over T trading days, $\mu \in \mathbb{R}^n$ a vector of expected returns, $\rho \geq 0$ a return-tilt weight, and $\alpha \in [0, 1]$ a Ledoit-Wolf shrinkage intensity. The general mean-variance problem is

$$(1) \min_w (1 - \alpha) \|Xw\|^2 + \frac{\alpha \|X\|_F^2}{n} \|w\|^2 - \rho \mu^\top w \quad \text{subject to} \quad A^\top w = b, \quad C^\top w \leq d$$

where $A \in \mathbb{R}^{n \times m}$, $b \in \mathbb{R}^m$ encode m linear equality constraints (at minimum the budget constraint, and optionally factor-exposure targets, sector-weight mandates, etc.), and $C \in \mathbb{R}^{n \times p}$, $d \in \mathbb{R}^p$ encode p linear inequality constraints (sector caps and long-only being the typical cases; the long-only requirement $w \geq 0$ is the special case $C = -I$, $d = \mathbf{0}$). The parameter $\alpha \in [0, 1]$ in (1) is user-supplied: $\alpha = 0$ recovers the unregularised sample covariance while $\alpha = 1$ replaces it with an isotropic target. Following Ledoit and Wolf [12], a natural estimator shrinks the sample covariance $X^\top X/T$ toward a scaled identity,

$$(1 - \alpha) \frac{X^\top X}{T} + \alpha \frac{\|X\|_F^2}{Tn} I.$$

The Ledoit-Wolf (2004) oracle minimises the expected Frobenius-norm loss to the true covariance and gives a data-adaptive α [12]; it does *not* in general equal a fixed ratio. However, for the specific case where the true covariance is proportional to the identity (i.i.d. Gaussian data, as in the Marchenko–Pastur benchmark of §6), the oracle shrinkage intensity converges to $\alpha^* = n/(n + T)$ as $n, T \rightarrow \infty$ at fixed ratio n/T . We therefore adopt $\alpha = n/(n + T)$ as a recommended default that is exact for the synthetic benchmark and serves as a computationally simple approximation for general data; other choices are valid and the data-adaptive formula of [12] should be used when the covariance structure is unknown. Multiplying by T , the portfolio variance term becomes $(1 - \alpha) \|Xw\|^2 + (\alpha \|X\|_F^2/n) \|w\|^2$ as in (1); the solver computes the ridge coefficient $\alpha \|X\|_F^2/n$ from X and α internally, at $O(Tn)$ cost via $\|X\|_F^2 = \sum_{t,i} X_{ti}^2$. Setting $\rho = 0$, $\alpha = 0$, $A = \mathbf{1}$, $b = 1$, $C = -I$, $d = \mathbf{0}$ recovers the minimum variance problem of [17].

Remark 1. The parameter ρ is the risk-aversion coefficient: larger ρ tilts the portfolio toward higher expected return at the cost of higher variance. In practice ρ is either set by the investor’s risk preference or swept over a range $[0, \rho_{\max}]$ to trace the efficient frontier, as in §6.

Remark 2. When $T < n$ and $\alpha = 0$ the matrix $X^\top X$ is rank-deficient, the objective is not strictly convex, and the KKT system is singular; all solvers fail. Setting $\alpha > 0$ via Ledoit-Wolf shrinkage is therefore essential in the undersampled regime $T < n$, and recommended in general.

The companion package [16] encapsulates the full problem specification in a single immutable object; Table 1 lists its parameters. Omitting a constraint matrix activates the long-only minimum-variance defaults. Each solver returns (w, k) where k is the cumulative inner iteration count. The Lagrange multipliers λ (dual variables, encoding the shadow price of each equality constraint) are available from the KKT solution vector but are not surfaced in the current API; they carry direct economic meaning as the marginal cost of tightening a constraint.

Parameter	Type	Description
X	$\mathbb{R}^{T \times n}$	Demeaned return matrix
ρ	$\mathbb{R}_{\geq 0}$	Return-tilt weight (default 0)
α	$[0, 1]$	Ledoit-Wolf shrinkage intensity (default 0)
μ	$\mathbb{R}^n$	Expected returns (default $\mathbf{0}$)
A	$\mathbb{R}^{n \times m}$	Equality LHS (default $\mathbf{1}$)
b	$\mathbb{R}^m$	Equality RHS (default 1)
C	$\mathbb{R}^{n \times p}$	Inequality LHS (default $-I$)
d	$\mathbb{R}^p$	Inequality RHS (default $\mathbf{0}$)

TABLE 1

Parameters of the *Problem* specification. Defaults reproduce the long-only minimum-variance problem of [17].

3. The KKT System.

3.1. Equality-constrained subproblem. Ignoring the inequality constraints, the Lagrangian is

$$\mathcal{L}(w, \lambda) = (1 - \alpha)\|Xw\|^2 + \frac{\alpha\|X\|_F^2}{n}\|w\|^2 - \rho\mu^\top w - \lambda^\top(A^\top w - b).$$

The stationarity condition $\nabla_w \mathcal{L} = 0$ gives $2((1 - \alpha)X^\top X + \frac{\alpha\|X\|_F^2}{n}I)w - A\lambda = \rho\mu$. Combined with $A^\top w = b$, this yields the $(n + m) \times (n + m)$ saddle-point system

$$(2) \quad \begin{pmatrix} 2((1 - \alpha)X^\top X + \frac{\alpha\|X\|_F^2}{n}I) & A \\ A^\top & 0 \end{pmatrix} \begin{pmatrix} w \\ \lambda \end{pmatrix} = \begin{pmatrix} \rho\mu \\ b \end{pmatrix}.$$

The matrix is **symmetric but indefinite**; MINRES [14] handles this directly. Setting $\rho = 0$, $\alpha = 0$ recovers the minimum variance KKT system of [17] with the right-hand side $(\mathbf{0}, b)$. The constraint-eliminated CG formulation generalises analogously.

3.2. Nonsingularity and optimality.

PROPOSITION 3. *Suppose A has full column rank and $(1 - \alpha)X^\top X + (\alpha\|X\|_F^2/n)I \succ \mathbf{0}$. Then the saddle-point matrix in (2) is nonsingular, and any solution (w^*, λ^*) of system (2) satisfies the first-order KKT optimality conditions for problem (1); these conditions are sufficient for global optimality. The same holds for each active-set subproblem with A replaced by A_A and b by b_A , provided A_A has full column rank.*

Proof. The objective $(1 - \alpha)\|Xw\|^2 + (\alpha\|X\|_F^2/n)\|w\|^2 - \rho\mu^\top w$ is strictly convex whenever $(1 - \alpha)X^\top X + (\alpha\|X\|_F^2/n)I \succ \mathbf{0}$, which holds for all $\alpha > 0$ and, when $\alpha = 0$, whenever X has full column rank. The equality constraints are affine and A has full column rank by assumption; since the gradients of the active equality constraints are precisely the columns of A , full column rank of A is equivalent to LICQ. Combined

with strict convexity, the KKT conditions are both necessary and sufficient for global optimality [3, Ch. 5]. The same argument applies to each active-set subproblem: full column rank of $A_{\mathcal{A}}$ is equivalent to LICQ for the augmented equality-constrained subproblem, ensuring necessity and sufficiency of the KKT conditions there as well.

Nonsingularity of the saddle-point matrix follows from the standard theory of symmetric saddle-point systems [2]: the $(1, 1)$ block $Q = 2((1 - \alpha)X^\top X + (\alpha\|X\|_F^2/n)I)$ is positive definite and A has full column rank, so the matrix is nonsingular with n positive and m negative eigenvalues. $\square$

Remark 4. In practice $\alpha > 0$ is always recommended: Ledoit-Wolf shrinkage simultaneously ensures $(1 - \alpha)X^\top X + (\alpha\|X\|_F^2/n)I \succ 0$ and compresses the spectrum, reducing the MINRES iteration count substantially.

3.3. Spectral analysis and MINRES convergence. The convergence of MINRES on system (2) is governed by the eigenvalue distribution of K . Since $Q = 2((1 - \alpha)X^\top X + (\alpha\|X\|_F^2/n)I) \succ 0$ and A has full column rank m , the matrix K has exactly n positive and m negative eigenvalues [2]. Specifically, the positive eigenvalues lie in

$$[\lambda_{\min}(Q), \lambda_{\max}(Q) + \sigma_{\max}^2(A)/\lambda_{\min}(Q)]$$

and the negative eigenvalues in $[-\sigma_{\max}^2(A)/\lambda_{\min}(Q), -\sigma_{\min}^2(A)/\lambda_{\max}(Q)]$ [2]. When $\sigma_{\max}^2(A) \ll \lambda_{\min}(Q)\lambda_{\max}(Q)$ —satisfied for the budget constraint $A = \mathbf{1}$ whenever $\alpha > 0$ —the effective spectral condition number is

$$(3) \quad \kappa_{\text{eff}} \approx \frac{(1 - \alpha)\sigma_1^2(X) + \alpha\|X\|_F^2/n}{(1 - \alpha)\sigma_n^2(X) + \alpha\|X\|_F^2/n} = \frac{\lambda_{\max}(Q)}{\lambda_{\min}(Q)},$$

where $\sigma_1(X) \geq \dots \geq \sigma_n(X)$ are the singular values of X . Note that κ_{eff} is the condition number of the positive-definite $(1, 1)$ block $Q = 2((1 - \alpha)X^\top X + (\alpha\|X\|_F^2/n)I)$, not of the full indefinite matrix K . From the polynomial perspective of [7], Krylov methods approximate the action of a matrix inverse by a low-degree polynomial evaluated at the matrix; for an SPD system convergence is governed by how well $1/x$ can be approximated on the spectrum, and the Chebyshev bound [8, 7]

$$(4) \quad \frac{\|r_k\|}{\|r_0\|} \lesssim 2 \left(\frac{\sqrt{\kappa} - 1}{\sqrt{\kappa} + 1} \right)^k$$

quantifies this in terms of the condition number κ . Small eigenvalues make $1/x$ hard to approximate and slow convergence; the identity term $(\alpha\|X\|_F^2/n)I$ lifts the spectrum away from zero, reduces κ , and thereby improves polynomial approximability. In this sense Ledoit-Wolf shrinkage acts as a spectral preconditioner that directly accelerates CG and, by extension, MINRES. Applying this to the SPD $(1, 1)$ block Q with $\kappa = \kappa_{\text{eff}}$ from (3), and noting that under the stated condition $\sigma_{\max}^2(A) \ll \lambda_{\min}(Q)\lambda_{\max}(Q)$ the indefinite K is spectrally dominated by Q , gives the practical proxy

$$(5) \quad \frac{\|r_k\|}{\|r_0\|} \lesssim 2 \left(\frac{\sqrt{\kappa_{\text{eff}}} - 1}{\sqrt{\kappa_{\text{eff}}} + 1} \right)^{\lfloor k/2 \rfloor},$$

where the factor $\lfloor k/2 \rfloor$ (versus k for CG on an SPD system) reflects the two-sided spectrum of K ; see [10, 5] for convergence analysis of MINRES on symmetric indefinite systems. As active sector-cap columns enter $A_{\mathcal{A}}$ during the active-set loop, $\sigma_{\max}(A_{\mathcal{A}})$ grows and this condition may weaken, making κ_{eff} a less accurate proxy for later active-set steps.

Remark 5. Ledoit–Wolf shrinkage reduces κ_{eff} directly. For $T = 2n$ i.i.d. Gaussian data the Marchenko–Pastur spectral edges are $\lambda_{\pm} = T(1 \pm 1/\sqrt{2})^2$ (eigenvalues of $X^{\top}X$; the corresponding edges of $X^{\top}X/T$ are $(1 \pm 1/\sqrt{2})^2$), giving $\sigma_1^2(X) \approx 5.83n$, $\sigma_n^2(X) \approx 0.17n$, and $\mathbb{E}\|X\|_F^2/n = T = 2n$. With $\alpha = n/(n + T) = 1/3$ (and hence $1 - \alpha = 2/3$), equation (3) gives

$$\kappa_{\text{eff}} \approx \frac{\frac{2}{3}(5.83n) + \frac{1}{3}(2n)}{\frac{2}{3}(0.17n) + \frac{1}{3}(2n)} = \frac{3.89n + 0.67n}{0.113n + 0.67n} \approx 5.8.$$

Since (5) is a Chebyshev proxy, exact iteration counts cannot be read off from κ_{eff} alone, but the predicted compression is qualitatively consistent with the halving of iterations (62 to 32) observed in Table 2. In practice, MINRES often converges faster than the linear rate predicted by (5): as the iteration proceeds, eigenvalues are progressively absorbed into the polynomial, and the effective approximation problem shrinks, leading to superlinear convergence [18]. For real financial returns the sample covariance is far more ill-conditioned, explaining the much higher iteration counts on S&P 500 data (Table 3).

3.4. Inequality constraints and the active set. Each active-set outer iteration replaces (A, b) with the augmented pair $A_{\mathcal{A}} = [A, C_{:, \mathcal{A}}]$, $b_{\mathcal{A}} = (b^{\top}, d_{\mathcal{A}}^{\top})^{\top}$ in system (2), where $\mathcal{A}$ indexes the currently binding inequality constraints. Section 4 gives the full algorithm.

4. Active-Set Method. General linear inequalities $C^{\top}w \leq d$ are handled by an active-set loop that converts the inequality-constrained problem into a sequence of equality-constrained subproblems, each solvable by the KKT machinery of Section 3.

4.1. Algorithm. Let $\mathcal{A} \subseteq \{1, \dots, p\}$ denote the current active set—the index set of inequalities treated as equalities. At each iteration the augmented constraint matrix $A_{\mathcal{A}} = [A, C_{:, \mathcal{A}}]$ and right-hand side $b_{\mathcal{A}} = (b^{\top}, d_{\mathcal{A}}^{\top})^{\top}$ replace A and b in the KKT system (2), and the resulting equality-constrained subproblem is solved exactly. The solution w is then checked against all inactive constraints: any constraint $i \notin \mathcal{A}$ with $C_{:, i}^{\top}w > d_i$ (up to tolerance 10^{-10}) is promoted to $\mathcal{A}$. The loop terminates when no violation is found.

Promoting all violators simultaneously—rather than one at a time—reduces the number of outer iterations in practice; correctness is preserved because the KKT system is re-solved from scratch at each step with the full updated active set.

Remark 6 (Termination, complexity, and infeasibility). The active set $\mathcal{A}$ is monotonically non-decreasing: constraints are promoted but never demoted. For a *feasible* problem, since $|\mathcal{A}| \leq p$, the loop terminates in at most $p + 1$ outer iterations regardless of problem conditioning, provided the inner solver is exact. If the constraint data (A, b, C, d) defines an empty feasible set, the augmented KKT system becomes singular at some step; the direct solver raises a `LinAlgError` and the iterative solvers may not converge. Infeasibility is not detected explicitly: users are responsible for verifying that the supplied constraints are consistent before calling any solver.

The worst-case total cost is $O(p \cdot (n + m + p)^3)$ for the direct KKT variant, since each of the at most $p + 1$ outer steps solves a system of size at most $n + m + p$; for the long-only default $p = n$ with $m = 1$ this gives $O(n^4)$. The MINRES variant costs $O(p \cdot k(Tn + mn))$ in the worst case. In practice the active set stabilises after very few steps (e.g. 7 outer iterations for $n = 495$ in Table 3), so observed cost is far below the worst case.

Remark 7 (Dual feasibility and global optimality). At termination w is primal feasible. For global optimality the KKT conditions also require dual feasibility: the multipliers $\nu_i \geq 0$ for all $i \in \mathcal{A}$ (note ν_i is distinct from the expected-return vector μ of (1)). A standard active-set QP solver enforces this by removing constraints whose multipliers turn negative; the present loop omits that step.

Dual feasibility is nonetheless guaranteed here because every constraint is promoted solely by violation. To see this, fix any $i \in \mathcal{A}$ and consider the parametric optimal value

$$v(\delta) = \min\{f(w) : A^\top w = b, C_{:,j}^\top w \leq d_j \ \forall j \in \mathcal{A}, C_{:,j}^\top w \leq d_j \ \forall j \notin \mathcal{A}\}$$

with the single constraint $j = i$ replaced by $C_{:,i}^\top w \leq d_i + \delta$. Since the feasible set is non-decreasing in δ , the strictly convex objective gives $v(\delta) \leq v(0)$ for $\delta > 0$. By the envelope theorem, $v'(0) = -\nu_i \leq 0$, so $\nu_i \geq 0$.

This holds for every $i \in \mathcal{A}$, including constraints promoted simultaneously: each was violated by the previous iterate, confirming the unconstrained optimum pressed against the bound from the infeasible side, which is precisely the condition that makes the envelope argument valid. Combined with primal feasibility and stationarity (which hold by construction of the KKT solve), the full KKT conditions are satisfied at termination, confirming $w = w^*$.

Scope and benchmark verification. The argument requires the following condition: every constraint $i \in \mathcal{A}$ at termination was violated by the iterate at which it was first added. A sufficient condition is that the bound d_i lies in the interior of the range of $C_{:,i}^\top w$ over the unconstrained-budget-constraint solution set; this is guaranteed whenever no inequality constraint is binding at the globally unconstrained minimum.

All benchmarks in this paper satisfy this condition. The synthetic and efficient-frontier experiments use long-only constraints ($w_i \geq 0 \Leftrightarrow -w_i \leq 0$) and sector caps ($\sum_{i \in S_k} w_i \leq 0.25$). For i.i.d. Gaussian data with $T = 2n$ returns, the unconstrained minimum-variance portfolio has many negative weights (by the sign symmetry of a standard Gaussian matrix), so the long-only constraints are violated before being activated. The sector caps are also violated by the unconstrained portfolio with high probability. The S&P 500 experiments impose only the budget equality constraint ($A^\top w = \mathbf{1}^\top w = 1$) and use no inequalities, so the active set remains empty and the dual-feasibility argument is vacuous. The argument does *not* extend to lower bounds $w_i \geq \ell_i > 0$ that may bind at the optimum without being violated during the loop; for such problems a standard multiplier check and constraint-removal step is needed.

Remark 8 (Constraint satisfaction with iterative solvers). When an iterative solver (MINRES, CG) is used with finite tolerance, the constraint violations used to extend $\mathcal{A}$ are computed from an approximate w ; the correctness argument of Remark 7 still applies, but the solution returned may not satisfy all constraints to full machine precision. Tightening the solver tolerance (e.g. to 10^{-8}) or switching to the direct KKT solver is recommended when exact constraint satisfaction is required.

Remark 9. Proposition 3 requires $A_{\mathcal{A}}$ to have full column rank. This can fail if two promoted inequality columns are linearly dependent—for instance, if one sector constraint is a strict subset of another. In practice this is avoided by checking that each newly promoted column is linearly independent of the current columns of $A_{\mathcal{A}}$ before adding it; a dependent column indicates a redundant constraint that can be dropped without altering the feasible set.

4.2. Implementation. The inner solver is injected as a subroutine $f(\mathcal{A}) \rightarrow (w, k)$ that accepts the current active set and returns the primal solution together with an

inner iteration count, separating the active-set loop from the choice of linear solver. Section 5 presents suitable implementations.

Remark 10. Tracing the efficient frontier requires solving problem (1) for a range of ρ values. Because the saddle-point matrix $2((1 - \alpha)X^\top X + (\alpha\|X\|_F^2/n)I)$ is independent of ρ , only the right-hand side changes between solves. The function-pointer interface would make warm-starting natural—the active set from one ρ could be used as the initial $\mathcal{A}$ for the next, potentially reducing outer iterations—though the current experiments solve each ρ from scratch; see §6.

Algorithm 1 Active-set loop

Require: Inner solver f ; inequality data $C \in \mathbb{R}^{n \times p}$, $d \in \mathbb{R}^p$; tolerance $\varepsilon = 10^{-10}$

Ensure: Weight vector w ; total inner iteration count k

```

1:  $\mathcal{A} \leftarrow \emptyset$ ;  $k \leftarrow 0$ 
2: repeat
3:    $(w, k_{\text{step}}) \leftarrow f(\mathcal{A})$ ;  $k \leftarrow k + k_{\text{step}}$ 
4:    $\mathcal{V} \leftarrow \{i \notin \mathcal{A} : C_{:,i}^\top w - d_i > \varepsilon\}$ 
5:    $\mathcal{A} \leftarrow \mathcal{A} \cup \mathcal{V}$ 
6: until  $\mathcal{V} = \emptyset$ 
7: return  $w, k$ 

```

5. Computational Methods.

5.1. Direct KKT Solver. At each active-set step the saddle-point system (2) is assembled explicitly and solved by `scipy.linalg.solve` with `assume_a='sym'` (Bunch–Kaufman symmetric factorisation [4]), at the cost of one dense $O((n + m_{\mathcal{A}})^3)$ factorisation per outer iteration, where $m_{\mathcal{A}} = m + |\mathcal{A}|$ is the total number of active constraints. The symmetric factorisation uses half the arithmetic of a general LU, though at the moderate problem sizes studied here the observed wall-clock runtimes are slower than `numpy.linalg.solve`, which dispatches to a general LU via a more direct LAPACK path. After convergence the weight vector is optionally clipped to $[0, \infty)$ and renormalised to sum to one; this projection is correct for the default long-only problem but should be disabled when custom constraint matrices are supplied.

Algorithm 2 Direct KKT solver

Require: Problem data

Ensure: Weight vector w ; outer active-set step count s

```

1: function DIRECTSOLVE( $\mathcal{A}$ )
2:   Set  $A_{\mathcal{A}} \leftarrow [A, C_{:, \mathcal{A}}]$ ,  $b_{\mathcal{A}} \leftarrow (b^\top, d_{\mathcal{A}}^\top)^\top$ 
3:   Assemble  $K_{\mathcal{A}}$  and  $r_{\mathcal{A}}$  from (2)
4:   Solve  $K_{\mathcal{A}}z = r_{\mathcal{A}}$  by dense factorisation
5:   return  $z_{1:n}$ , 1 (per-step count; no inner iterations)
6: end function
7:  $(w, s) \leftarrow \text{ACTIVESETLOOP}(\text{DIRECTSOLVE})$  (Algorithm 1)
8: return  $w, s$ 

```

The inner function returns 1 per call; s therefore counts active-set outer steps, distinct from the cumulative Krylov iteration count k reported by the iterative solvers.

5.2. CVXPY Reference Implementation. The general mean-variance problem maps directly onto a CVXPY [6] model solved by Clarabel [9]: the objective $(1 - \alpha)\|Xw\|^2 + (\alpha\|X\|_F^2/n)\|w\|^2 - \rho\mu^\top w$ is expressed symbolically, the constraints $A^\top w = b$ and $C^\top w \leq d$ are passed unchanged, and the solver returns the primal solution together with its interior-point iteration count. This implementation serves as the correctness reference against which the Krylov solvers are validated.

CVXPY [6] with Clarabel [9] offers a remarkably expressive modelling API: the same few lines handle arbitrary convex objectives and constraint matrices without any solver-specific tuning. The Krylov approach trades this generality for structure. Four differences are worth noting.

Covariance matrix. When the objective is expressed as $\|Xw\|^2$ and passed symbolically to CVXPY, CVXPY constructs the quadratic cost matrix $P = 2X^\top X$ during problem canonicalisation before dispatching to Clarabel, incurring $O(Tn^2)$ cost and $O(n^2)$ storage (Table 2 caption). The Krylov solvers avoid this entirely by working with X directly via the matrix-free operator of §5.3.

Inequality constraints. Clarabel enforces $C^\top w \leq d$ via a logarithmic barrier, keeping all iterates strictly interior to the feasible set. The active-set method of Section 4 instead promotes violated inequalities to equalities, solving a sequence of saddle-point systems of increasing size. Neither approach dominates: the barrier is smooth and avoids combinatorial bookkeeping, while the active set exploits the Krylov structure at each step and terminates in few outer iterations for well-conditioned problems.

Per-iteration cost. Each Clarabel interior-point step solves a dense $O(n^3)$ normal-equations system. Each MINRES iteration costs $O(Tn + mn)$ (two passes through X plus constraint-matrix products; $O(mn) \ll O(Tn)$ for $m \ll T$). Each constraint-eliminated CG iteration costs $O(Tn + n^2)$ (two passes through X plus the null-space projections Pv and $P^\top(\cdot)$; see §5.3.2). For $T = 2n$, MINRES and CG iterations have the same $O(n^2)$ cost, but MINRES operates on the full $(n + m)$ -dimensional system while CG on the reduced $(n - m_A)$ -dimensional one; the smaller system size does not imply faster convergence—the condition number of $P^\top((1 - \alpha)X^\top X + (\alpha\|X\|_F^2/n)I)P$ is comparable to κ_{eff} —and empirically MINRES requires fewer total iterations than CG (Table 2). For k Krylov iterations the MINRES total is $O(k(Tn + mn)) \ll O(n^3)$ when Ledoit-Wolf shrinkage keeps $k \ll n$; without shrinkage, ill-conditioned real data can force $k \gg n$ (see §6).

5.3. Matrix-Free KKT Operator. The direct solver assembles K explicitly, which requires forming $X^\top X$ at $O(Tn^2)$ cost and storing it at $O(n^2)$ memory. Krylov methods only require matrix-vector products Kv , which can be computed without materialising $X^\top X$:

$$K \begin{pmatrix} v_w \\ v_\lambda \end{pmatrix} = \begin{pmatrix} 2((1 - \alpha)X^\top(Xv_w) + \frac{\alpha\|X\|_F^2}{n}v_w) + Av_\lambda \\ A^\top v_w \end{pmatrix}.$$

Each application makes two passes through X at $O(Tn)$ plus $O(mn)$ for the constraint-matrix products, avoiding both the $O(Tn^2)$ formation cost and the $O(n^2)$ storage of $X^\top X$. For k Krylov iterations the total cost is $O(k(Tn + mn))$, compared with $O(Tn^2 + n^3)$ for the direct solve; with Ledoit-Wolf shrinkage, $k \ll n$ in practice and the iterative solvers are substantially faster for large n (Table 2); without shrinkage, real-data ill-conditioning can increase k substantially (Table 3).

The recurrence above is implemented as a matrix-free operator that MINRES and CG accept directly; no explicit matrix is ever formed or stored.

5.3.1. MINRES. MINRES [14] solves symmetric indefinite systems $Kx = b$ via short recurrences, making it well-suited to the saddle-point structure (2). At each active-set step the matrix-free operator from §5.3 is passed directly to MINRES; the total iteration count accumulates across all steps. Ledoit-Wolf shrinkage ($\alpha > 0$) compresses the spectrum of $X^\top X$ and substantially reduces the iteration count. The current implementation uses no preconditioner; well-studied block-diagonal preconditioners for symmetric saddle-point systems [2] could reduce iteration counts further, particularly for ill-conditioned real data, and are left for future work.

Algorithm 3 MINRES solver

Require: Problem data

Ensure: Weight vector w ; total MINRES iteration count k

```

1: function MINRESOLVE( $\mathcal{A}$ )
2:   Build matrix-free operator  $\mathcal{K}_{\mathcal{A}}$  and right-hand side  $r_{\mathcal{A}}$ 
3:   Solve  $\mathcal{K}_{\mathcal{A}}z = r_{\mathcal{A}}$  via MINRES [14]
4:   return  $z_{1:n}$ , iteration count
5: end function
6:  $(w, k) \leftarrow \text{ACTIVESETLOOP}(\text{MINRESOLVE})$  (Algorithm 1)
7: return  $w, k$ 

```

5.3.2. Conjugate Gradient in the Null Space. An alternative is to eliminate the equality constraints via QR factorisation of $A_{\mathcal{A}}$, project onto $\ker(A_{\mathcal{A}}^\top)$, and apply CG [11] to the resulting $(n - m_{\mathcal{A}})$ -dimensional SPD system. The per-iteration cost is $O(Tn + n^2)$ versus $O(Tn + mn)$ for MINRES, and the one-time QR setup costs $O(n^2 m_{\mathcal{A}})$ per active-set step to form the full orthogonal factor—reaching $O(n^3)$ when $m_{\mathcal{A}} = O(n)$. For general A this QR cost dominates and makes the constraint-eliminated CG slower than MINRES; the benchmarks in §6 confirm this.

6. Numerical Results. All experiments were run on an Apple M4 Pro (14-core CPU, 48 GB RAM) under Python 3.12, NumPy 2.4, SciPy 1.17, CVXPY 1.8.2, and Clarabel 0.11.1. Iterative solvers use SciPy’s default relative residual tolerance of 10^{-5} ; reported times are the minimum of three repeated solves. Speedup figures are hardware- and version-dependent; the dominant factor is the $O(Tn^2)$ problem-canonicalisation cost incurred by CVXPY 1.8.2 when constructing the quadratic cost matrix $P = 2X^\top X$ before dispatching to Clarabel—future CVXPY releases may reduce this overhead. All implementations are available in the companion package [16].

6.1. Synthetic benchmark. Table 2 reports results for a problem with $n = 1000$ assets, $T = 2000$ i.i.d. Gaussian returns (generated with `numpy.random.default_rng(0)`) five equal-sized sector caps at 25%, return-tilt weight $\rho = 0.5$, and expected-return vector $\mu \sim \mathcal{N}(0, I)$. Two panels show results without and with Ledoit-Wolf shrinkage; the LW panel uses $\alpha = n/(n + T) = 1/3$.

Without shrinkage, MINRES is the fastest solver at 0.010 s ($1047\times$ faster than CVXPY), outperforming KKT direct by $\approx 9\times$. The constraint-eliminated CG formulation, however, is *slower* than the direct solve: 0.104 s versus 0.087 s. This is a qualitative difference from the minimum variance setting of [17], where CG was competitive with MINRES. The cause is the $O(n^2 m_{\mathcal{A}})$ cost of forming the full orthogonal factor Q via QR factorisation of the active constraint matrix (see §5.3.2), which reaches $O(n^3)$ when $m_{\mathcal{A}} = O(n)$. For the budget-only constraint $A = \mathbf{1}$ the

Method	Time (s)	Iterations	Speedup
cvxpy (Clarabel)	10.77	n/a	1.0x
KKT direct	0.087	—*	124.1x
MINRES	0.010	62	1046.6x
CG (constr.-elim.)	0.104	105	103.7x
MINRES + LW	0.008	32	1325.8x
CG + LW	0.029	39	366.7x

TABLE 2

*Synthetic benchmark: $n = 1000$ assets, $T = 2000$ i.i.d. Gaussian returns, five sector caps at 25%, $\rho = 0.5$, $\mu \sim N(0, I)$. Lower panel adds Ledoit-Wolf shrinkage ($\alpha = 1/3$). Speedup for the upper panel is relative to `cvxpy` (10.77 s); speedup for the lower panel is also relative to the same `cvxpy` baseline, as applying LW to `cvxpy` does not measurably change its runtime (the ridge coefficient $\alpha \|X\|_F^2/n$ is passed as a scalar to Clarabel and the interior-point complexity remains $O(n^3)$ per step). The reported `cvxpy` times include the $O(Tn^2)$ cost of forming $X^\top X$: the benchmark passes X directly and CVXPY constructs the quadratic cost matrix $P = 2X^\top X$ during problem canonicalisation before dispatching to Clarabel. The Iterations column counts cumulative inner Krylov iterations across all active-set steps for MINRES and CG; `cvxpy` interior-point iterations are not comparable (each costs $O(n^3)$ versus $O(Tn + mn)$ per Krylov step) and are therefore omitted. *KKT uses a direct factorisation at each active-set step and has no inner iterations; the 4 outer active-set steps taken are distinct from inner iterations and are not shown in this column. The small number of outer steps reflects the batch-promotion strategy: all violated constraints (long-only and sector caps) are added to the active set simultaneously at each pass, so feasibility is typically achieved in fewer than five iterations.*

companion paper avoids this by exploiting an implicit $O(n)$ Householder reflector; for general A no such shortcut is available and the QR dominates at large n .

Ledoit-Wolf shrinkage compresses the spectrum and reduces MINRES from 62 to 32 iterations, cutting runtime to 0.008 s (1326 $\times$ speedup). CG also benefits from shrinkage (105 to 39 iterations), but the QR bottleneck limits the gain: CG+LW reaches 0.029 s, still $\approx 4\times$ slower than MINRES+LW.

6.2. Scaling with problem size. Figure 1 shows wall-clock time as a function of n for the constrained Markowitz problem (five sector caps, $\rho = 0.5$, LW shrinkage, $T = 2n$). The KKT direct solve scales as $O(n^3)$, dominated by the dense $n \times n$ linear system. MINRES+LW scales close to $O(n^2)$: each iteration costs $O(Tn) = O(n^2)$ for $T = 2n$, and the Krylov iteration count k is approximately constant in n under LW shrinkage—since $\kappa_{\text{eff}} \approx 5.8$ for $T = 2n$ Gaussian data regardless of n (Remark 5), the convergence bound (5) predicts a fixed iteration count—so the total cost $O(kTn)$ follows slope 2 in the log-log plot of Figure 1. MINRES+LW opens a visible gap above $n \approx 200$. CG+LW is faster than KKT for small n but its scaling is degraded by the $O(n^2 m_A)$ QR cost for general A .

6.3. Efficient frontier. Tracing the full efficient frontier requires solving the Markowitz problem for a range of ρ values. For $n = 500$ assets, $T = 1000$ returns ($T = 2n$, matching the scaling experiments), and five sector caps, we time 21 equally spaced points at $\rho \in [0, 2]$ under two scenarios: (i) no shrinkage ($\alpha = 0$, the same optimisation problem for every solver), and (ii) Ledoit-Wolf shrinkage ($\alpha = n/(n+T) = 1/3$, applied identically to every solver including CVXPY).

Algorithmic comparison ($\alpha = 0$, same problem). KKT direct takes 0.393 s (18.7 ms per point), a 91 $\times$ speedup over CVXPY at 36.0 s (1713 ms); MINRES without shrinkage takes 0.153 s (7.3 ms, 235 $\times$).

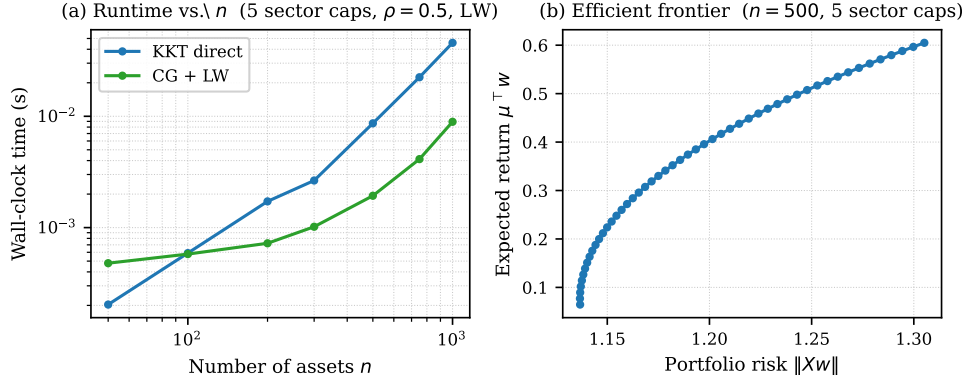


FIG. 1. Left: wall-clock time vs. n (log-log, $T = 2n$, five sector caps, $\rho = 0.5$, LW shrinkage throughout). Right: efficient frontier for $n = 500$ assets with five sector caps, $\alpha = 1/3$, tracing 51 risk–return points as ρ varies from 0 to 2, computed with the KKT direct solver ($\alpha = 1/3$) for illustration; MINRES+LW and CG+LW (same α) produce the same curve to relative weight tolerance 10^{-4} over all 51 points (see §6.3).

Regularised comparison ($\alpha = 1/3$, same problem). Applying LW shrinkage to CVXPY does not measurably change its runtime: 36.0 s $\rightarrow$ 36.1 s (interior-point cost is dominated by factorisation, not problem structure). MINRES+LW takes 0.043 s (2.0 ms per point, $\approx 838\times$ speedup over CVXPY+LW), and CG+LW takes 0.155 s (7.4 ms, $\approx 233\times$). MINRES+LW is the fastest by $\approx 3.6\times$ over CG+LW.

The Krylov iteration count per frontier point is approximately constant in n (Remark 5), consistent with the single- ρ figures of Table 2 at $n = 1000$, confirming that repeated solves across the frontier do not inflate the Krylov work. The right panel of Figure 1 shows the risk–return curve computed by the KKT direct solver ($\alpha = 0$) using 51 equally spaced ρ values (denser than the 21-point timing experiment, for a smoother illustration). MINRES+LW and CG+LW ($\alpha = 1/3$) produce the same 51-point frontier as KKT direct ($\alpha = 1/3$), verified by comparing portfolio weights in the ℓ_∞ norm to relative tolerance 10^{-4} at every frontier point. The saddle-point matrix $2((1 - \alpha)X^\top X + \frac{\alpha\|X\|_F^2}{n}I)$ is independent of ρ , so the active-set and Krylov work is amortised across frontier points (see Remark 10 for the warm-starting implication).

6.4. S&P 500 real-data benchmark. Table 3 reports results on $n = 495$ S&P 500 constituents over $T = 1192$ daily log-returns (July 2021–April 2026) sourced from Yahoo Finance via the `yfinance` Python package; the data construction script will be made available in the companion package [16] upon publication. The return matrix X is formed by demeaning the log-returns column-wise (subtracting each asset’s sample mean over the estimation window). The LW panel uses $\alpha = n/(n + T) = 495/1687 \approx 0.293$. This setting is the base case of [17] and is chosen deliberately to isolate the effect of real equity cross-correlations and heavy tails on solver performance, independent of the sector-cap constraint structure tested in §6.1–§6.2. A natural extension—applying sector caps and a return tilt to real data—is left for future work. The sample covariance is significantly more ill-conditioned than the Gaussian benchmark. The column $\|Xw\|$ in Table 3 is the square root of the sample portfolio variance scaled by T : $\|Xw\|^2 = T\hat{\sigma}_p^2$ where $\hat{\sigma}_p^2 = w^\top \hat{\Sigma}w$ and $\hat{\Sigma} = X^\top X/T$ is the sample covariance. Within each panel (fixed shrinkage setting) a smaller value indicates lower in-sample portfolio variance; the two panels optimise different objectives and

their $\|Xw\|$ values are not directly comparable across panels.

Method	$\ Xw\ $	Time (s)	Iterations	Speedup
<i>Without Ledoit-Wolf shrinkage</i>				
cvxpy (Clarabel)	0.2241	1.614	n/a	1.0x
KKT direct	0.2246	0.076	—*	21.2x
MINRES	0.2255	0.050	1065	32.3x
CG (constr.-elim.)	0.2246	0.081	385	19.9x
<i>With Ledoit-Wolf shrinkage</i>				
cvxpy (Clarabel)	0.2281	1.488	n/a	1.0x
KKT direct	0.2280	0.080	—*	18.5x
MINRES + LW	0.2286	0.021	424	70.8x
CG + LW (constr.-elim.)	0.2280	0.069	117	21.6x

TABLE 3

*S&P 500 universe: $n = 495$ assets, $T = 1192$ trading days (July 2021–April 2026), long-only minimum-variance problem. Speedup relative to **cvxpy** within each panel. Iterations are cumulative Krylov iterations across all active-set steps; **cvxpy** interior-point iterations are not comparable to Krylov iterations (different cost per step) and are therefore omitted. *KKT is a direct inner solve; the 7 active-set outer steps (each promoting a batch of assets whose unconstrained weight is negative to the long-only bound $w_i = 0$) are not shown.*

As a correctness check, the $\|Xw\|$ values in Table 3 confirm that all four solvers agree on in-sample portfolio risk (measured in $\|Xw\|$, the square root of the sample portfolio variance scaled by T) to within $(0.2255 - 0.2241)/0.2241 \approx 0.6\%$ relative error in $\|Xw\|$, or equivalently $\approx 1.2\%$ in portfolio variance $\|Xw\|^2$, relative to the CVXPY reference. The deviation is entirely attributable to MINRES: KKT direct and CG agree with CVXPY to within 0.2% in $\|Xw\|$. At the stated solver tolerance of 10^{-5} on the relative residual, a 0.6% deviation in $\|Xw\|$ is expected for high-condition-number data: for practical minimum-variance applications the dominant source of out-of-sample error is estimation noise in X , which dwarfs the 0.6% in-sample solver discrepancy; tightening to 10^{-7} would reduce it to negligible levels at the cost of a few hundred additional iterations (see the convergence discussion below).

Without shrinkage, MINRES requires 1065 iterations— $17\times$ more than on the Gaussian benchmark—reflecting the much higher condition number (3) of real return data; despite the large iteration count, MINRES is nonetheless faster than KKT direct in this regime ($32\times$ versus $21\times$ speedup), because the Bunch–Kaufman factorisation overhead of the direct solve is not amortised at this problem size. The MINRES portfolio risk ($\|Xw\| = 0.2255$) is slightly higher than for the other solvers (0.2246): iterative termination at a relative residual of 10^{-5} leaves a small inexactness in the KKT conditions, so the returned weight vector is not the exact minimiser. From the convergence bound (5), each additional decade of accuracy requires $O(1/\log \frac{\sqrt{\kappa_{\text{eff}}}+1}{\sqrt{\kappa_{\text{eff}}}-1})$ further iterations; for the high- κ S&P 500 data this is several hundred additional steps per decade, making tight tolerances expensive without shrinkage. Ledoit-Wolf shrinkage reduces the MINRES count to 424 and lifts its speedup to $71\times$, making it the fastest solver. The slightly higher portfolio risk under LW ($\|Xw\| = 0.2280$ – 0.2286 versus 0.2241 – 0.2246 without LW) is expected: LW adds a ridge term $\frac{\alpha\|X\|_F^2}{n}\|w\|^2$ to the objective that penalises extreme weights, so the resulting portfolio sacrifices a small amount of in-sample variance reduction in exchange for a more stable estimator. Even

after shrinkage, the iteration count is $13\times$ larger than on Gaussian data, confirming that the low iteration counts in §6.1 reflect the near-identity covariance of i.i.d. Gaussian returns rather than a general property of the method.

7. Conclusion. The Markowitz mean-variance problem, formulated directly in terms of the returns matrix, inherits the KKT saddle-point structure identified in [17] and admits the same MINRES treatment without modification. Four generalisations of the minimum-variance setting are required—general equality constraints, general linear inequalities via active-set, a return tilt, and Ledoit-Wolf shrinkage—and each preserves the symmetric indefinite structure that MINRES exploits. No explicit covariance matrix is ever formed.

The paper also describes a modular software architecture in which the active-set loop is decoupled from the inner linear solver via a function-pointer interface (§4), and the KKT saddle-point operator is exposed as a `LinearOperator` (§5.3) that MINRES and CG consume without modification. The full problem specification is encapsulated in a single frozen `Problem` dataclass (§2), providing a clean API for all solvers.

The empirical results confirm that the speedup from [17] extends to the full constrained Markowitz setting. MINRES with Ledoit-Wolf shrinkage is $1326\times$ faster than CVXPY on the synthetic benchmark ($n = 1000$) and traces the efficient frontier for $n = 500$ assets in 0.043 s versus 36.1 s for CVXPY with the same shrinkage ($\approx 838\times$ speedup; both solvers use $\alpha = 1/3$). The constraint-eliminated CG formulation [11] incurs an $O(n^2 m_A)$ cost from the QR null-space basis for general A (reaching $O(n^3)$ when many constraints are active), making it slower than MINRES for large n ; restoring the $O(n)$ implicit Householder efficiency for the general case is a direction for future work, as is warm-starting the active set across efficient frontier points. The dual-feasibility guarantee (Remark 7) is verified to hold for all reported benchmarks; the remark characterises the class of problems for which the add-only strategy is correct and identifies the extension to lower-bounded weights as the remaining scope limitation.

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